An Architecture for a Network Anomaly Detection Framework
draft-ietf-nmop-network-anomaly-architecture-08
draft-ietf-nmop-network-anomaly-semantics-06
draft-ietf-nmop-network-anomaly-lifecycle-06

Motivation and architecture of a Network Anomaly Detection Framework
and the relationships to other documents describing
network symptom semantics and network incident lifecycle

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Problem Statement and Motivation

How it is being addressed in each document

Network Anomaly Detection

To **detect service interruptions** before users do, network operations must use **automated**, **holistic** monitoring across all **three network planes**.

- draft-ietf-nmop-network-anomaly-architecture describes the **motivation**, **architecture** and the **relationship** to other two documents.

By **standardizing labeled incident data** and **automating the postmortem process**, operators can **improve machine learning** models continuously and **foster better collaboration** between vendors and academia.

- draft-ietf-nmop-network-anomaly-semantics defines Symptom semantics to enable **standardized data exchange** to **validate results** with network engineers and improve supervised and semi-supervised machine learning systems.
- draft-ietf-nmop-network-anomaly-lifecycle describes the lifecycle process, enabling network engineers to **interact** with the network anomaly detection system to **refine the detection abilities** over time.

Network Anomaly Detection Architecture

DRAFT-IETF-NMOP-NETWORK-ANOMALY-ARCHITECTURE-08

Network Anomaly Detection Architecture

Document summary

Motivation:

Human-driven troubleshooting does not scale due to the growing number of dependencies

Approach:

This draft defines a standardized, architecture-level framework

Outcome:

Enable automated service disruption detection

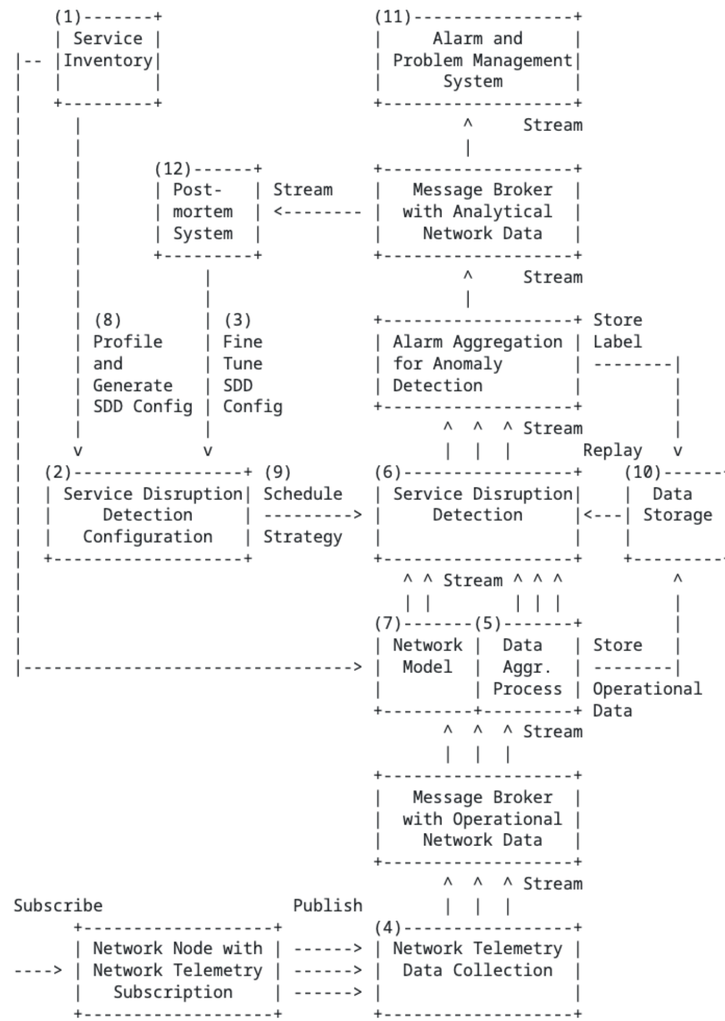

Figure 1: Service Disruption Detection System Architecture

draft-ietf-nmop-network-anomaly-architecture

Network Anomaly Detection Architecture

Key draft updates since v-07, IETF 125

- **Changes in References**
  - SIMAP documents downstream references, draft-ietf-nmop-simap-concept and draft-havel-nmop-simap-yang, changed from normative to informative.
  - Knowledge Graph downstream references, draft-mackey-nmop-kg-for-netops, changed from normative to informative.
- **Network Incident Contribution**
  - Updated Network Anomaly Detection relationship in draft-ietf-nmop-network-incident-yang-10#section-9.5

Semantic Metadata Annotation

DRAFT-IETF-NMOP-NETWORK-ANOMALY-SEMANTICS-06

Semantic Metadata Annotation

Document summary

Motivation:

Enable the exchange of network anomaly data across diverse systems and support benchmarking of anomaly detection solutions

Approach:

Define a standardized semantic metadata schema and a set of semantically structured terms

Outcome:

Provide human-readable context for describing and understanding network anomalies

Action	Reason	Trigger
Interface State	Up	Link-Layer
Interface State	Down	Link-Layer
Interface Statistics	Errors	-
Interface Statistics	Discards	-
Interface Statistics	Unknown Protocol	-

Table 3: Description of symptoms and their actions, reasons and triggers for Management Plane.

Semantic Metadata Annotation

Key draft updates since v-06, IETF 125

- **Abstract**
  - Simplified.
- **Schema change**
  - Term "cbl" removed from YANG module and grouping names.
  - Updated YANG prefix names.
- **Examples**
  - Added relevant state instance data example draft-ietf-nmop-network-anomaly-semantics-06#section-4.4.2.
- **ietf-network-anomaly-service-topology**
  - Detailed on reasoning why this YANG module is needed.
  - At the IETF 125 ONSEN discussions (slides-125-onsen-proposed-charter-discussion), the point on that the existing L2 and L3 service models lack certain attributes and map ability to SIMAP resp. RFC 8345 was raised. Therefore, therefore opted for having dedicated service and network groupings in ietf-network-anomaly-service-topology. This document is classified experimental. We expect that the IETF NMOP, ONSEN is advancing in these areas so that ietf-network-anomaly-service-topology no longer is needed.

```
module: ietf-network-anomaly-service-topology

  augment /rsn:relevant-state/rsn:service:
    +--:(l2vpn)
    |  +--rw l2vpn-service* [vpn-id]
    |     +--rw vpn-id                string
    |     +--rw uri?                  inet:uri
    |     +--rw vpn-name?             string
    |     +--rw site-ids*             string
    |     +--rw change-id?            yang:uuid
    |     +--rw change-start-time?    yang:date-and-time
    |     +--rw change-end-time?      yang:date-and-time
    +--:(l3vpn)
       +--rw l3vpn-service* [vpn-id]
          +--rw vpn-id                string
          +--rw uri?                  inet:uri
          +--rw vpn-name?             string
          +--rw site-ids*             string
          +--rw change-id?            yang:uuid
          +--rw change-start-time?    yang:date-and-time
          +--rw change-end-time?      yang:date-and-time
  augment /rsn:relevant-state-notification/rsn:service:
    +--:(l2vpn)
    |  +-- l2vpn-service* [vpn-id]
    |     +-- vpn-id                string
    |     +-- uri?                  inet:uri
    |     +-- vpn-name?             string
    |     +-- site-ids*             string
    |     +-- change-id?            yang:uuid
    |     +-- change-start-time?    yang:date-and-time
    |     +-- change-end-time?      yang:date-and-time
    +--:(l3vpn)
       +-- l3vpn-service* [vpn-id]
          +-- vpn-id                string
          +-- uri?                  inet:uri
          +-- vpn-name?             string
          +-- site-ids*             string
          +-- change-id?            yang:uuid
          +-- change-start-time?    yang:date-and-time
          +-- change-end-time?      yang:date-and-time
  augment /rsn:relevant-state/rsn:anomaly:
    +--rw vpn-node-terminations* [hostname vrf-name]
       +--rw hostname              inet:host
       +--rw vrf-id?               uint32
       +--rw vrf-name              string
       +--rw route-distinguisher?  string
       +--rw interface-id*         uint32
       +--rw interface-name*       string
       +--rw peer-ip*              inet:ip-address
       +--rw next-hop*             inet:ip-address
  augment /rsn:relevant-state-notification/rsn:anomaly:
    +-- vpn-node-terminations* [hostname vrf-name]
       +-- hostname              inet:host
       +-- vrf-id?               uint32
       +-- vrf-name              string
       +-- route-distinguisher?  string
       +-- interface-id*         uint32
       +-- interface-name*       string
       +-- peer-ip*              inet:ip-address
       +-- next-hop*             inet:ip-address
```

Network Anomaly Lifecycle

DRAFT-IETF-NMOP-NETWORK-ANOMALY-LIFECYCLE-06

Network Anomaly Lifecycle

Document summary

Motivation:

Bridge the gap between automated network anomaly detection and actionable operational tasks

Approach:

Define a standardized, iterative lifecycle for Network anomaly

Outcome:

Continuously Improve the correctness of anomaly detection through human-in-the-loop feedback

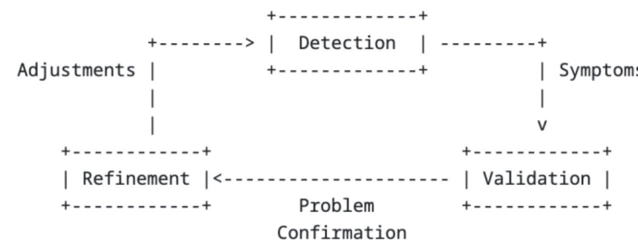

Figure 1: Anomaly Detection Refinement Lifecycle

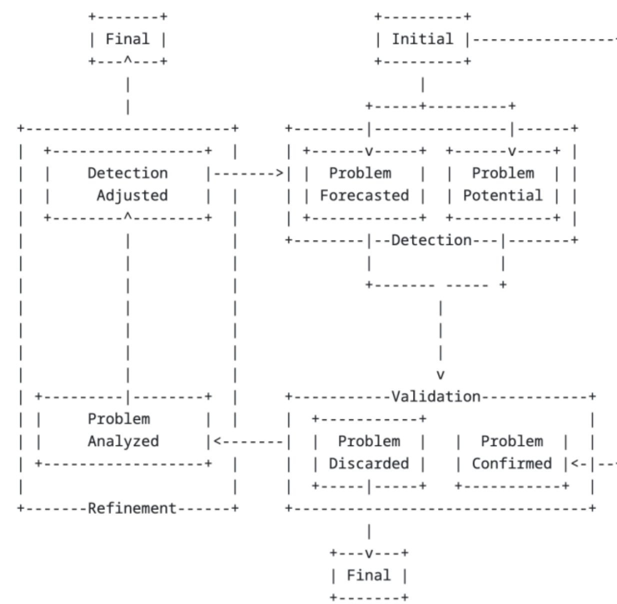

Figure 2: Network Anomaly State Machine

Network Anomaly Lifecycle

Key draft updates since v-05, IETF 125

- **Abstract and Introduction**
  - Refactored and simplified abstraction and introduction sections.
  - Make better use of abstract and introduction section by moving the reasoning from abstract to introduction.
- **Lifecycle Process**
  - Detailed on when lifecycle process when it changes from validation to refinement stage.
- **Schema change**
  - " topic-name " and " subject-name " leaves have now their own message broker typedefs aligning with draft-ietf-nmop-yang-message-broker-message-key

```
module: ietf-relevant-state
  +--rw relevant-state!
     +--rw id                     yang:uuid
     +--rw uri?                   inet:uri
     +--rw description?           string
     +--rw start-time             yang:date-and-time
     +--rw end-time?              yang:date-and-time
     +--rw strategy?              string
     +--rw confidence-score?      score
     +--rw concern-score          score
     +--rw stage                  identityref
     +--rw (service)?
     +--rw anomaly* [id revision]
        +--rw id                     yang:uuid
        +--rw revision               yang:counter32
        +--rw uri?                   inet:uri
        +--rw stage                  identityref
        +--rw description?           string
        +--rw start-time             yang:date-and-time
        +--rw end-time?              yang:date-and-time
        +--rw confidence-score?      score
        +--rw pattern?               identityref
        +--rw annotator
        |  +--rw id?                  yang:uuid
        |  +--rw name                 string
        |  +--rw version?             string
        |  +--rw annotator-type?      enumeration
        +--rw operational-data
        |  +--rw topic-name?        message-broker-topic
        |  +--rw subject-name?      message-broker-subject
        +--rw symptom!
           +--rw id                  yang:uuid
           +--rw concern-score       score
```

Next Steps and Remaining Issues

Feedback on latest changes

Next Steps

- Request YANG doctors review for draft-ietf-nmop-network-anomaly-semantics-06 and draft-ietf-nmop-network-anomaly-lifecycle-06.
- Extend the postmortem system requirements, accompanied by an implementation PoC

Remaining Issues

- No known remaining issues at this time.